

HDOC - DAY 12

C Programming Activity

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SUBJECT: C programming
DATE: 5-09026

Question 1

Find the sum and product of the first and last digits of a positive integer. Example: Number = 12345, Sum = 6, Product = 5.

1. Aim

To write, compile, execute, and test a menu-driven C program that finds the sum and product of the first and last digits of a positive integer.

2. Algorithm

1. Start the program.
2. Declare the required integer variables.
3. Input a positive integer n.
4. Display the menu and input the user choice.
5. For choice 1, find the last digit using $n \% 10$.
6. Copy n into temp and repeatedly divide temp by 10 until only the first digit remains.
7. Calculate $\text{sum} = \text{first} + \text{last}$ and $\text{product} = \text{first} * \text{last}$.
8. Display the sum and product.
9. Stop the program.

3. Test Case Table

Test Case	Input Number	Choice	Expected Output	Result
TC1	12345	1	Sum = 6, Product = 5	Pass
TC2	9876	1	Sum = 15, Product = 54	Pass
TC3	505	1	Sum = 10, Product = 25	Pass

4. C Program

```
#include <stdio.h>
int main() {
    int n, choice, temp, first, last, sum, product;
```

```

int k, digit, lesser = 0, equal = 0, greater = 0;

printf("Enter a positive integer: ");
scanf("%d", &n);
if (n <= 0) {
    printf("Please enter a positive integer.\n");
    return 0;
}

printf("\n1. Sum and product of first and last digits\n");
printf("2. Count digits lesser than, equal to, and greater than k\n");
printf("Enter your choice: ");
scanf("%d", &choice);

switch (choice) {
    case 1:
        last = n % 10;
        temp = n;
        while (temp >= 10)
            temp = temp / 10;
        first = temp;
        sum = first + last;
        product = first * last;
        printf("Sum = %d\n", sum);
        printf("Product = %d\n", product);
        break;

    case 2:
        printf("Enter digit k: ");
        scanf("%d", &k);
        if (k < 0 || k > 9) {
            printf("k must be a digit from 0 to 9.\n");
            break;
        }
        temp = n;
        while (temp > 0) {
            digit = temp % 10;
            if (digit < k)
                lesser++;
            else if (digit == k)
                equal++;
            else
                greater++;
            temp = temp / 10;
        }
        printf("Count_lesser = %d\n", lesser);
        printf("Count_equal = %d\n", equal);
        printf("Count_greater = %d\n", greater);
        break;

    default:
        printf("Invalid choice.\n");
}
return 0;
}

```

5. Write / Save the Program

Open the terminal in Kali Linux and create the source file:

```
nano day12.c
```

Paste the C program shown above. Press Ctrl+O, Enter, then Ctrl+X to save and exit.

6. Compile the Program

```
gcc day12.c -o day12
```

If no error is displayed, the program has compiled successfully.

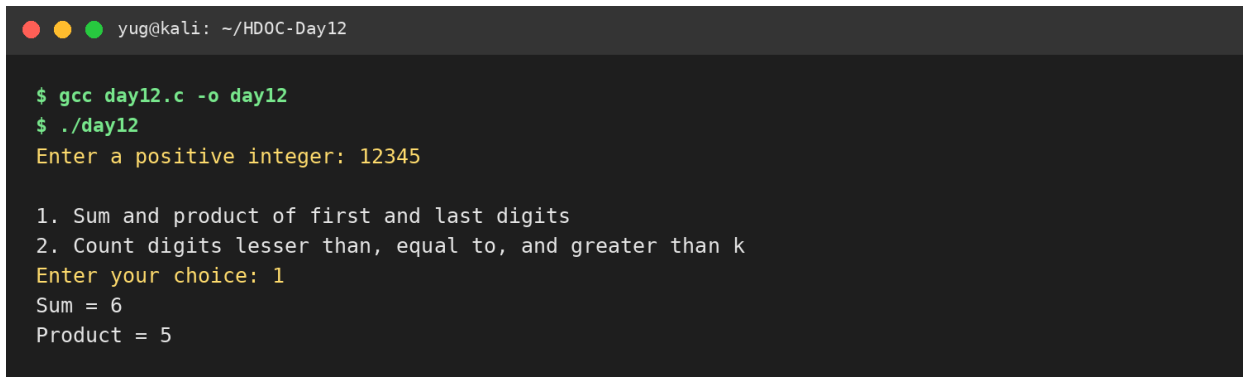
7. Execute the Program

```
./day12
```

8. Test Against the Prepared Use Cases

Run the program for each prepared test case, select the required menu choice, and compare the actual output with the expected output.

Execution Example 1

A terminal window with a dark background and light-colored text. The prompt is 'yug@kali: ~/HD0C-Day12'. The user enters '\$ gcc day12.c -o day12', followed by '\$./day12'. The program prompts 'Enter a positive integer: 12345'. The user enters '1'. The program displays two menu options: '1. Sum and product of first and last digits' and '2. Count digits lesser than, equal to, and greater than k'. The user enters '1'. The program outputs 'Sum = 6' and 'Product = 5'.

```
yug@kali: ~/HD0C-Day12  
  
$ gcc day12.c -o day12  
$ ./day12  
Enter a positive integer: 12345  
  
1. Sum and product of first and last digits  
2. Count digits lesser than, equal to, and greater than k  
Enter your choice: 1  
Sum = 6  
Product = 5
```

Execution Example 2

```
yug@kali: ~/HD0C-Day12

$ gcc day12.c -o day12
$ ./day12
Enter a positive integer: 9876

1. Sum and product of first and last digits
2. Count digits lesser than, equal to, and greater than k
Enter your choice: 1
Sum = 15
Product = 54
```

Execution Example 3

```
yug@kali: ~/HD0C-Day12

$ gcc day12.c -o day12
$ ./day12
Enter a positive integer: 505

1. Sum and product of first and last digits
2. Count digits lesser than, equal to, and greater than k
Enter your choice: 1
Sum = 10
Product = 25
```

The three execution examples above match the expected results in the prepared test case table.

Question 2

Count how many digits are lesser than, equal to, and greater than a given digit 'k'. Example: Number = 12345, k = 3, Count_lesser = 2, Count_equal = 1, Count_greater = 2.

1. Aim

To write, compile, execute, and test a menu-driven C program that compares every digit of a positive integer with a given digit k and counts digits that are lesser than, equal to, or greater than k.

2. Algorithm

1. Start the program.
2. Declare integer variables and initialize lesser, equal, and greater counters to 0.
3. Input a positive integer n.
4. Display the menu and input the user choice.
5. For choice 2, input digit k (0 to 9).
6. Copy n into temp.
7. Repeat while temp > 0: obtain digit = temp % 10.
8. If digit < k, increment lesser; if digit == k, increment equal; otherwise increment greater.
9. Remove the last digit using temp = temp / 10.
10. Display Count_lesser, Count_equal, and Count_greater.
11. Stop the program.

3. Test Case Table

Test Case	Input Number	k	Choice	Expected L,E,G	Result
TC1	12345	3	2	2, 1, 2	Pass
TC2	55555	5	2	0, 5, 0	Pass
TC3	2468	5	2	2, 0, 2	Pass

4. C Program

```
#include <stdio.h>
int main() {
    int n, choice, temp, first, last, sum, product;
    int k, digit, lesser = 0, equal = 0, greater = 0;

    printf("Enter a positive integer: ");
    scanf("%d", &n);
    if (n <= 0) {
        printf("Please enter a positive integer.\n");
        return 0;
    }
}
```

```

printf("\n1. Sum and product of first and last digits\n");
printf("2. Count digits lesser than, equal to, and greater than k\n");
printf("Enter your choice: ");
scanf("%d", &choice);

switch (choice) {
    case 1:
        last = n % 10;
        temp = n;
        while (temp >= 10)
            temp = temp / 10;
        first = temp;
        sum = first + last;
        product = first * last;
        printf("Sum = %d\n", sum);
        printf("Product = %d\n", product);
        break;

    case 2:
        printf("Enter digit k: ");
        scanf("%d", &k);
        if (k < 0 || k > 9) {
            printf("k must be a digit from 0 to 9.\n");
            break;
        }
        temp = n;
        while (temp > 0) {
            digit = temp % 10;
            if (digit < k)
                lesser++;
            else if (digit == k)
                equal++;
            else
                greater++;
            temp = temp / 10;
        }
        printf("Count_lesser = %d\n", lesser);
        printf("Count_equal = %d\n", equal);
        printf("Count_greater = %d\n", greater);
        break;

    default:
        printf("Invalid choice.\n");
}
return 0;
}

```

5. Write / Save the Program

Open the terminal in Kali Linux and create the source file:

```
nano day12.c
```

Paste the C program shown above. Press Ctrl+O, Enter, then Ctrl+X to save and exit.

6. Compile the Program

```
gcc day12.c -o day12
```

If no error is displayed, the program has compiled successfully.

7. Execute the Program

```
./day12
```

8. Test Against the Prepared Use Cases

Run the program for each prepared test case, select the required menu choice, and compare the actual output with the expected output.

Execution Example 1

```
yug@kali: ~/HD0C-Day12

$ gcc day12.c -o day12
$ ./day12
Enter a positive integer: 12345

1. Sum and product of first and last digits
2. Count digits lesser than, equal to, and greater than k
Enter your choice: 2
Enter digit k: 3
Count_lesser = 2
Count_equal = 1
Count_greater = 2
```

Execution Example 2

```
yug@kali: ~/HD0C-Day12

$ gcc day12.c -o day12
$ ./day12
Enter a positive integer: 55555

1. Sum and product of first and last digits
2. Count digits lesser than, equal to, and greater than k
Enter your choice: 2
Enter digit k: 5
Count_lesser = 0
Count_equal = 5
Count_greater = 0
```

Execution Example 3

```
yug@kali: ~/HD0C-Day12

$ gcc day12.c -o day12
$ ./day12
Enter a positive integer: 2468

1. Sum and product of first and last digits
2. Count digits lesser than, equal to, and greater than k
Enter your choice: 2
Enter digit k: 5
Count_lesser = 2
Count_equal = 0
Count_greater = 2
```

The three execution examples above match the expected results in the prepared test case table.

Result

The menu-driven C program was written, compiled, executed, and tested against the prepared use cases. Both digit operations produced the expected results for the tested inputs.